



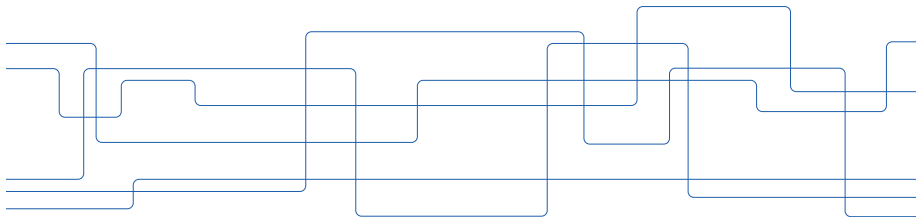
Conformal Prediction: Uncertainty Quantification and Drift Detection

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Outline

P-values

Nonconformity scores

Conformal predictors

- Conformal classifiers

- Conformal regressors

- Conformal predictive systems

- Mondrian conformal predictors

Conformal test martingales

How (un)certain is the prediction?

- ▶ A single class label or regression value that is output by a model does not indicate the extent to which we can rely on it
- ▶ Techniques for **uncertainty quantification** can express the prediction uncertainty in various ways, e.g., using class probability distributions or prediction intervals
- ▶ Many uncertainty quantification techniques rely on distributional assumptions and/or target specific model types, while provided guarantees are often asymptotic
- ▶ **Conformal prediction** is a model-agnostic framework for uncertainty quantification, which provides finite-sample guarantees under a rather weak assumption (*exchangeability*)

A *p*-variable P is a random variable that satisfies:

$$\mathbb{P}(P \leq \epsilon) \leq \epsilon, \forall \epsilon \in (0, 1) \quad (1)$$

The values taken by a *p*-variable are *p-values*

Let $\alpha_1, \alpha_2, \dots$ denote an *exchangeable* sequence of values where each $\alpha_i \in \mathbb{R}$

A *p-value* p_n for α_n is computed by:

$$p_n = \frac{|i : \alpha_i > \alpha_n| + \tau_n |i : \alpha_i = \alpha_n|}{n} \quad (2)$$

where $i \in \{1, \dots, n\}$ and $\tau_n \sim \mathcal{U}(0, 1)$

The p-values $p_1, p_2, \dots$ are IID and distributed uniformly on $[0, 1]$

Nonconformity scores

Let $z_1, z_2, \dots$ denote an exchangeable sequence of examples, where each $z_i \in \mathcal{Z}$, for some example space $\mathcal{Z}$

Let $\alpha_1 = A(z_1)$, $\alpha_2 = A(z_2)$, $\dots$ denote the sequence of *nonconformity scores*, obtained using some *nonconformity function*:

$$A : \mathcal{Z} \rightarrow \mathbb{R}$$

Conformal predictors

Let $z_i = (\mathbf{x}_i, y_i)$, where $\mathbf{x}_i \in \mathcal{X}$, for some object space $\mathcal{X}$, and $y_i \in \mathcal{Y}$ for some label space $\mathcal{Y}$, where $\mathcal{Y} = \{y_1, \dots, y_k\}$ for classification and $\mathcal{Y} = \mathbb{R}$ for regression

Given a calibration set $Z_{cal} = \{z_1, \dots, z_q\}$, a nonconformity function A , a significance level ϵ , and a test object $\mathbf{x}_{q+1}$, a *conformal predictor* outputs a *prediction set*:

$$\Gamma^\epsilon = \{y \in \mathcal{Y} : z_{q+1} = (\mathbf{x}_{q+1}, y) \ \& \ p_{q+1} \geq \epsilon\} \quad (3)$$

This gives the following (exact) coverage guarantee:

$$\mathbb{P}(y_{q+1} \in \Gamma^\epsilon) = 1 - \epsilon \quad (4)$$

Nonconformity scores for classifiers

The nonconformity scores can be computed using a model (h) that has been obtained from a separate training set:

$$A(z = (\mathbf{x}, y)) = A_h(\mathbf{x}, y)$$

Two common nonconformity functions for conformal classifiers are:

$$A_h(\mathbf{x}, y) = 1 - h(\mathbf{x}, y) \quad (\text{Hinge})$$

$$A_h(\mathbf{x}, y) = \max\{h(\mathbf{x}, y') : y' \in \mathcal{Y} \setminus \{y\}\} - h(\mathbf{x}, y) \quad (\text{Margin})$$

Conformal classification

	y	$h(\mathbf{x}, y)$	α
1	a	0.7	0.3
2	b	0.6	0.4
3	b	0.8	0.2
4	a	0.2	0.8
5	c	0.7	0.3
6	b	0.8	0.2
7	c	0.5	0.5
8	a	0.6	0.4
9	c	0.7	0.3

$$h(\mathbf{x}_{10}, a) = 0.6$$

$$h(\mathbf{x}_{10}, b) = 0.3$$

$$h(\mathbf{x}_{10}, c) = 0.1$$

$$A_h(\mathbf{x}_{10}, a) = 0.4 \quad p_{10} = (2 + 3\tau)/10$$

$$A_h(\mathbf{x}_{10}, b) = 0.7 \quad p_{10} = (1 + 1\tau)/10$$

$$A_h(\mathbf{x}_{10}, c) = 0.9 \quad p_{10} = (0 + 1\tau)/10$$

$$\Gamma^{0.1} = \{a, b\}$$

$$\Gamma^{0.2} = \{a\}$$

Nonconformity scores for regressors

Two common nonconformity functions for conformal regressors are:

$$A_h(\mathbf{x}, y) = |y - h(\mathbf{x})| \quad (\textit{Standard})$$

$$A_h(\mathbf{x}, y) = \frac{|y - h(\mathbf{x})|}{\sigma(\mathbf{x})} \quad (\textit{Normalized})$$

where σ is a function to estimate the difficulty of $\mathbf{x}$

Obtaining prediction intervals

To avoid testing infinitely number of labels for a test object $\mathbf{x}_n$, the following is employed for the standard nonconformity function:

$$\Gamma^\epsilon = h(\mathbf{x}_{q+1}) \pm \alpha'_s$$

where $\alpha'_1, \dots, \alpha'_q$ are the nonconformity scores in decreasing order and $s = \lfloor \epsilon(q+1) \rfloor$

Similarly, for the normalized nonconformity function:

$$\Gamma^\epsilon = h(\mathbf{x}_{q+1}) \pm \alpha'_s \sigma(\mathbf{x}_{q+1})$$

Instead of the exact coverage guarantee, we get a lower bound:

$$\mathbb{P}(y_{q+1} \in \Gamma^\epsilon) \geq 1 - \epsilon \quad (5)$$

Standard conformal regressors

	y	$h(\mathbf{x})$	α	α'
1	25	24	1	4
2	23	24	1	3
3	27	24	3	3
4	18	22	4	2
5	12	14	2	2
6	33	34	1	1
7	19	17	2	1
8	11	14	3	1
9	14	14	0	0

$$\Gamma^{0.1} = h(\mathbf{x}_{10}) \pm \alpha'_1 = h(\mathbf{x}_{10}) \pm 4$$

$$\Gamma^{0.2} = h(\mathbf{x}_{10}) \pm \alpha'_2 = h(\mathbf{x}_{10}) \pm 3$$

Normalized conformal regressors

	y	$h(\mathbf{x})$	$\sigma(\mathbf{x})$	α	α'
1	25	24	1	1	1.33
2	23	24	2	0.5	1
3	27	24	3	1	1
4	18	22	3	1.33	1
5	12	14	2	1	1
6	33	34	1	1	1
7	19	17	2	1	1
8	11	14	3	1	0.5
9	14	14	1	0	0

$$\Gamma^{0.1} = h(\mathbf{x}_{10}) \pm \alpha'_1 \sigma(\mathbf{x}_{10}) = h(\mathbf{x}_{10}) \pm 1.33 \sigma(\mathbf{x}_{10})$$

$$\Gamma^{0.2} = h(\mathbf{x}_{10}) \pm \alpha'_2 \sigma(\mathbf{x}_{10}) = h(\mathbf{x}_{10}) \pm \sigma(\mathbf{x}_{10})$$

Conformal predictive systems

Compute nonconformity scores by:

$$\alpha = \frac{y - h(\mathbf{x})}{\sigma(\mathbf{x})}$$

For a test object $\mathbf{x}_{q+1}$:

- let $C_0 = -\infty$ and $C_{q+1} = \infty$
- let $C_i = h(\mathbf{x}_{q+1}) + \alpha'_i \sigma(\mathbf{x}_{q+1})$ for $i \in \{1, \dots, q\}$
- output the conformal predictive distribution:

$$Q(y) = \begin{cases} \frac{i + \tau_{q+1}}{q+1} & \text{if } y \in (C_i, C_{i+1}) \text{ for } i \in \{0, \dots, q\} \\ \frac{i' - 1 + (i'' - i' + 2)\tau_{q+1}}{q+1} & \text{if } y = C_i \text{ for } i \in \{1, \dots, q\} \end{cases}$$

Conformal predictive systems

	y	$h(\mathbf{x})$	$\sigma(\mathbf{x})$	α	α'
1	25	24	1	1	-1.33
2	23	24	2	-0.5	-1
3	27	24	3	1	-1
4	18	22	3	-1.33	-1
5	12	14	2	-1	-0.5
6	33	34	1	-1	0
7	19	17	2	1	1
8	11	14	3	-1	1
9	14	14	1	0	1

$$C_1 = h(\mathbf{x}_{10}) + \alpha'_1 \sigma(\mathbf{x}), \dots, C_9 = h(\mathbf{x}_{10}) + \alpha'_9 \sigma(\mathbf{x})$$

Conformal predictive systems

For $h(\mathbf{x}_{10}) = 20$ and $\sigma_{10} = 1$:

$$C_0, \dots, C_{10} = -\infty, 18.67, 19, 19, 19, 19.5, 20, 21, 21, 21, \infty$$

$$Q(15) = \tau_{10}/10 \text{ since } 15 \in (C_0, C_1)$$

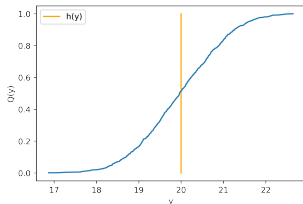
$$Q(18.8) = (1 + \tau_{10})/10 \text{ since } 18.8 \in (C_1, C_2)$$

$$Q(19) = (2 - 1 + (4 - 2 + 2)\tau_{10})/10 = (1 + 4\tau_{10})/10 \text{ since } 19 = C_2$$

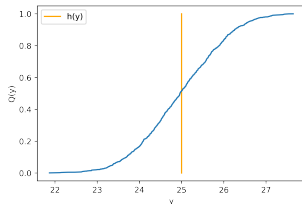
$$Q(22) = (9 + \tau_{10})/10 \text{ since } 22 \in (C_9, C_{10})$$

Conformal predictive distributions

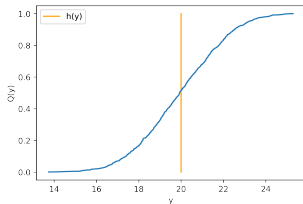
$$h(x) = 20 \quad \sigma = 1$$



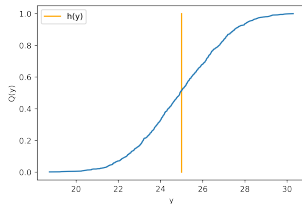
$$h(x) = 25 \quad \sigma = 1$$



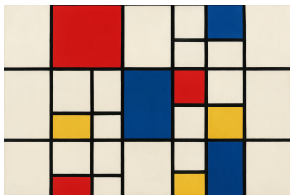
$$h(x) = 20 \quad \sigma = 2$$



$$h(x) = 25 \quad \sigma = 2$$



Mondrian conformal predictors

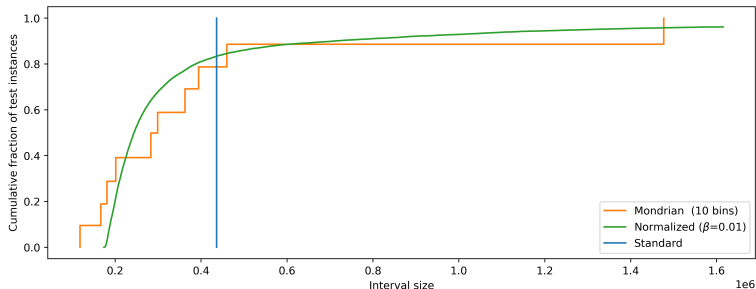


A *Mondrian conformal predictor* divides the examples into disjoint subsets according to a *Mondrian* taxonomy, and forms a regular conformal predictor from each subset

For conformal classifiers, the class labels may be used to form the categories; this allows for controlling the class-wise error rate

For conformal regressors and predictive systems, the categories can be formed by binning the difficulty estimates

Mondrian conformal regressors



Interval sizes for the house_sales dataset

Conformal test martingales

A *conformal test martingale* is computed by:

$$s_n = f_1(p_1) \cdots f_n(p_n) \quad (6)$$

where each f_i is a *betting function* $f : [0, 1] \rightarrow [0, \infty]$ satisfying:

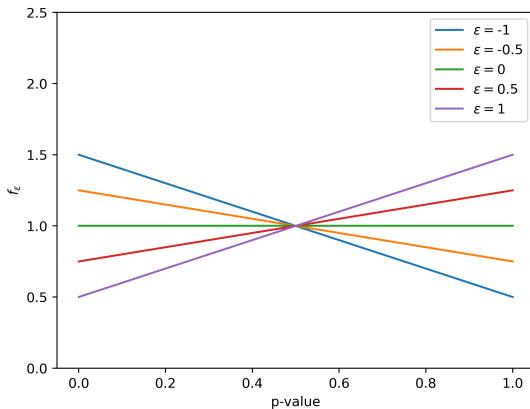
$$\int_0^1 f(u) du = 1 \quad (7)$$

If the sequence of examples $z_1, z_2, \dots$ is exchangeable, then:

$$\mathbb{P}(\{(p_1, p_2, \dots) : \exists n : s_n \geq c\}) \leq 1/c \quad (8)$$

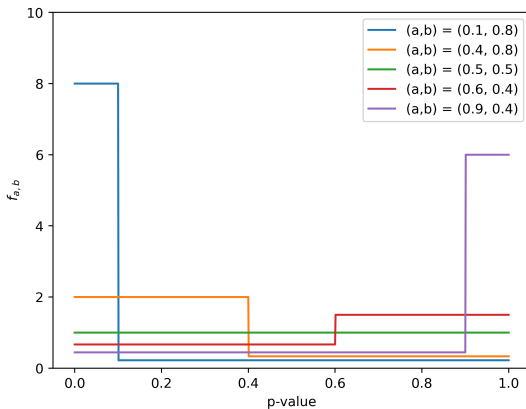
Betting functions

$$f_{\epsilon}(p) = 1 + \epsilon(p - 0.5)$$



Betting functions

$$f_{a,b}(p) = \begin{cases} \frac{b}{a} & \text{if } p \leq a \\ \frac{1-b}{1-a} & \text{otherwise} \end{cases}$$



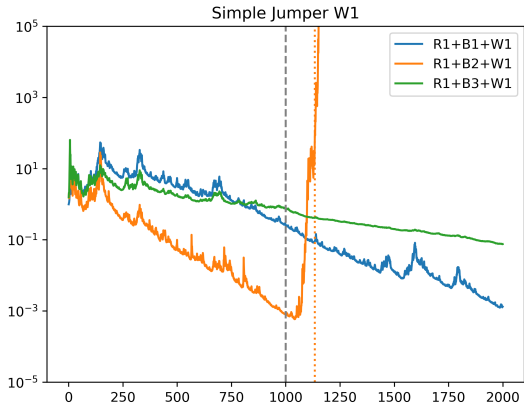
Algorithm 1 Simple Jumper

Input: $p_1, p_2, \dots$ (p-values), $t \in (0, 1)$ (tax rate), E (epsilons)

Output: $s_1, s_2, \dots$

```
1: for  $\epsilon \in E$  do  $c_\epsilon \leftarrow 1/|E|$ 
2: for  $i = 1, 2, \dots$  do
3:   for  $\epsilon \in E$  do  $c_\epsilon \leftarrow c_\epsilon f_\epsilon(p_i)$ 
4:    $s_i \leftarrow \sum_{\epsilon \in E} c_\epsilon$ 
5:   for  $\epsilon \in E$  do  $c_\epsilon \leftarrow (1 - t) c_\epsilon + (t/|E|) s_i$ 
```

Test case: $\mathcal{N}(0, 1) \rightarrow \mathcal{N}(0, 2)$



The Python package *crepes*



Conformal Classifiers, Regressors and Predictive Systems

pypi package 0.9.0 conda-forge 0.9.0 downloads 127k docs passing license BSD-3-clause release date october

<https://github.com/henrikbostrom/crepes>

<https://crepes.readthedocs.io>

Concluding remarks

- ▶ Conformal prediction provides means for controlling the probability of error; the framework is model-agnostic and relies on no other assumptions than that of exchangeability
- ▶ Conformal classifiers output sets of class labels, conformal regressors output prediction intervals, and conformal predictive systems output cumulative distribution functions
- ▶ Mondrian conformal predictors can be used to control the error across arbitrarily defined categories
- ▶ Conformal test martingales allow for detecting violations of the exchangeability assumption while controlling the false alarm rate



Some areas that we have not covered

- ▶ Transductive (online) approaches
- ▶ Multitarget problems
- ▶ Handling violations of the exchangeability assumption

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